# INTRODUCTION TO PROGRAMMING FOR DATA ANALYSIS PROJECT

EDENAKPO Owokoto Gildas & ELEKOUN Faizath

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```
Attachement du package : 'dplyr'

Les objets suivants sont masqués depuis 'package:stats':

    filter, lag

Les objets suivants sont masqués depuis 'package:base':

    intersect, setdiff, setequal, union

Warning: le package 'patchwork' a été compilé avec la version R 4.5.2


Please cite as:

 Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary Statistics Tables.

 R package version 5.2.3. https://CRAN.R-project.org/package=stargazer

Warning: le package 'xtable' a été compilé avec la version R 4.5.2

Warning: le package 'texreg' a été compilé avec la version R 4.5.2

Version:  1.39.4
Date:     2024-07-23
Author:   Philip Leifeld (University of Manchester)

Consider submitting praise using the praise or praise_interactive functions.
Please cite the JSS article in your publications -- see citation("texreg").


Attachement du package : 'texreg'

L'objet suivant est masqué depuis 'package:tidyr':

    extract
```

# 1 Import data

```
 [1] "DATAFLOW"    "LAST.UPDATE" "freq"        "unit"        "sex"
 [6] "indic_he"    "geo"         "TIME_PERIOD" "OBS_VALUE"   "OBS_FLAG"
[11] "CONF_STATUS"
```

```
 [1] "DATAFLOW"    "LAST.UPDATE" "freq"        "age"         "sex"
 [6] "unit"        "geo"         "TIME_PERIOD" "OBS_VALUE"   "OBS_FLAG"
[11] "CONF_STATUS"
```

```
  TIME_PERIOD     geo OBS_VALUE
1        2013 Albania        NA
2        2014 Albania        NA
3        2015 Albania        NA
4        2016 Albania        NA
5        2017 Albania      7.47
6        2018 Albania      6.98
```

```
  TIME_PERIOD     geo OBS_VALUE
1        2014 Austria      57.7
2        2015 Austria      58.0
3        2016 Austria      57.0
4        2017 Austria      57.1
5        2018 Austria      56.9
6        2019 Austria      57.3
```

```
[1] "TIME_PERIOD" "geo"         "OBS_VALUE"
```

```
[1] "TIME_PERIOD" "geo"         "OBS_VALUE"
```

```
  year country healthy_life_years
1 2014 Austria               57.7
2 2015 Austria               58.0
3 2016 Austria               57.0
4 2017 Austria               57.1
5 2018 Austria               56.9
6 2019 Austria               57.3
```

```
  year country inequality_income_dist
1 2013 Albania                     NA
2 2014 Albania                     NA
3 2015 Albania                     NA
4 2016 Albania                     NA
5 2017 Albania                   7.47
6 2018 Albania                   6.98
```

# 2 Description of the relationship of interest

Our data come from Eurostat and cover the period from 2014 to 2023 across European countries. ## Description of the variable X and Y

#### 2.0.1 Variable X : Inequality of income distribution (S80/S20 ratio)

The inequality of income distribution is expressed through the S80/S20 income quintile share ratio, which compares the total income received by the 20% of the population with the highest income (the richest quintile) to that received by the 20% with the lowest income (the poorest quintile). This indicator is a ratio without units, meaning it is not expressed in percentages. It provides a clear measure of how unevenly income is distributed among individuals in a society. A lower ratio (close to 1) indicates a more equal income distribution, while a higher ratio reflects greater inequality, where the richest earn many times more than the poorest. For example, a ratio of 4 means that the top 20% of earners receive four times the income of the bottom 20%.

#### 2.0.2 Variable Y : Healthy life years

We know people are living longer. However, do we live longer and better or do we gain only years of life in bad health?

The indicator of healthy life years (HLY) measures the number of remaining years that a person of specific age is expected to live without any severe or moderate health problems. The notion of health problem for Eurostat's HLY is reflecting a disability dimension and is based on a self-perceived question which aims to measure the extent of any limitations, for at least six months, because of a health problem that may have affected respondents as regards activities they usually do (the so-called GALI - Global Activity Limitation Instrument foreseen in the annual EU-SILC survey). The indicator is therefor also called disability-free life expectancy (DFLE). So, HLY is a composite indicator that combines mortality data with health status data.

HLY also monitor health as a productive or economic factor. An increase in healthy life years is one of the main goals for European health policy. And it would not only improve the situation of individuals but also result in lower levels of public health care expenditure. If healthy life years are increasing more rapidly than life expectancy, it means that people are living more years in better health.

### 2.1 Interest of studying the relationship between Inequality of income distribution and Healthy life years

Studying the relationship between income inequality and healthy life years is important because socioeconomic factors have a major influence on health outcomes. Higher income inequality can lead to disparities in access to healthcare, education, nutrition, and living conditions, which may reduce the number of years people live in good health. Understanding this relationship helps policymakers design more effective strategies to promote social equity and public health, ultimately improving overall well-being and reducing health inequalities within the population. Moreover, this type of analysis highlights the close links between economic conditions and well-being, and it aligns fully with the logic of sustainable development. It directly contributes to the United Nations Sustainable Development Goals (SDGs), particularly Goal 10 (Reduced Inequalities) and Goal 3 (Good Health and Well-being).

## 3 Description of our data

### 3.1 Measure of X

The variable X (inequality of income distribution) is measured by the S80/S20 income quintile share ratio. This indicator compares the total income received by the 20% of the population with the highest income to that received by the 20% with the lowest income. The ratio is a dimensionless number (not expressed in percentages). The data are typically collected and published by Eurostat, based on household income surveys conducted in each country.

### 3.2 Measure of Y

The variable Y (healthy life years) is measured as the average number of years a person can expect to live in good health, free from serious illness or disability. It is based on two main components: life expectancy and self-perceived health status, often gathered through population health surveys. The indicator is expressed in years, and data are available from Eurostat.

### 3.3 The columns used to merge both datasets X and Y

The two datasets are merged using the columns as both variables are measured annually and reported for each European country. These common identifiers allow for an accurate match between income inequality data and healthy life years data for the same country and year.

### 3.4 The unit of observation in the final dataset

The unit of observation in the final dataset is a country-year pair. Each row represents a specific country in a specific year (for example, France 2019 or Germany 2021).

### 3.5 The number of observation in the final dataset

The final dataset contains 330 observations, corresponding to the number of country-year combinations available after merging both datasets.

## 4 The goal of the project

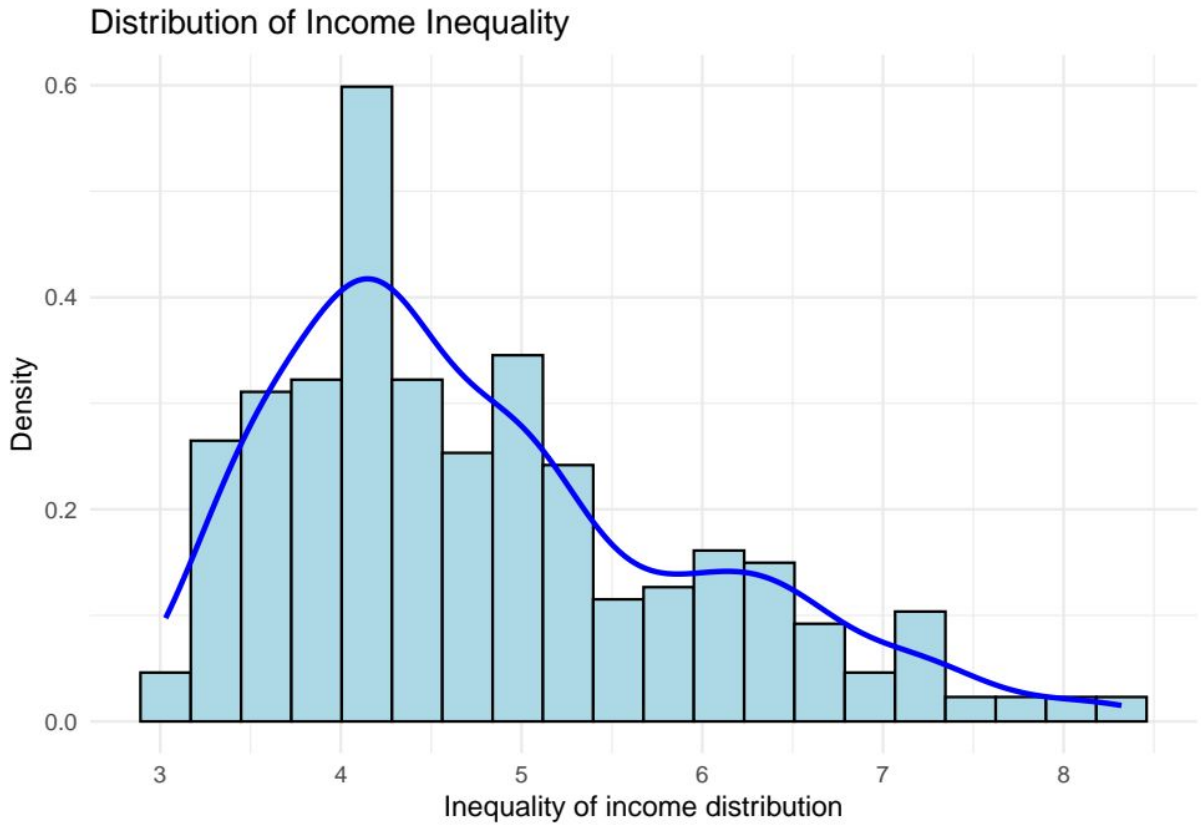
In summary, this project aims to study the relationship between income inequality and healthy life years across European countries over the period 2014 to 2023. The purpose is to better understand how economic disparities can influence health and well-being within societies.

## 5 Descriptive statistics

### 5.1 The distribution of the variable “inequality of income distribution”

### 5.2 Histogram and density plot of the variable “inequality of income distribution”

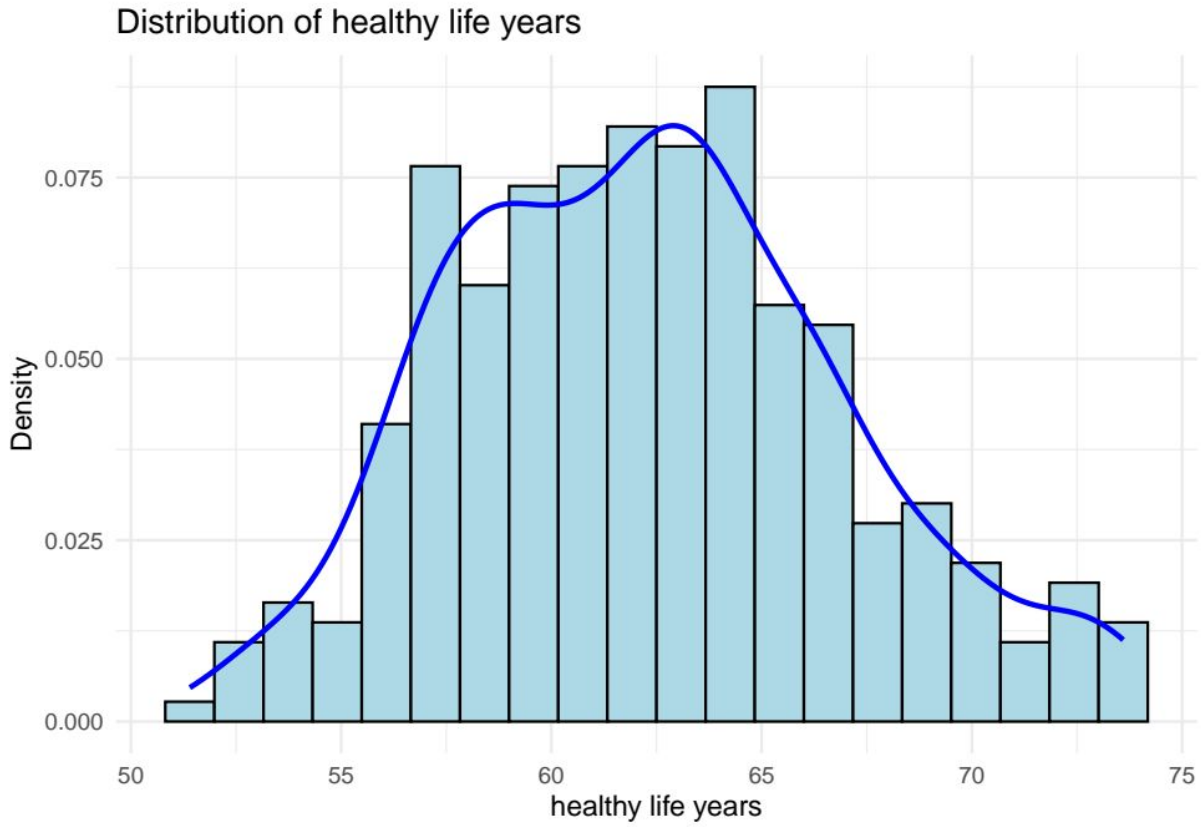
```
Warning: The dot-dot notation (‘..density..‘) was deprecated in ggplot2 3.4.0.
i Please use ‘after_stat(density)‘ instead.
This warning is displayed once every 8 hours.
Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
generated.
```




The histogram shows a right-skewed distribution, with a main peak around the value 4, indicating that most observations correspond to a moderate level of income inequality. The frequency gradually decreases for higher values, revealing the presence of a few cases of very high inequality, but these are relatively rare. This pattern suggests that, across the countries studied, income distribution is fairly balanced for most, while a small number experience significant income disparities.

### 5.3 Distribution of the variable "Healthy life years"

### 5.4 Histogram and density plot of the variable "Healthy life years"




The histogram of healthy life years displays a roughly symmetrical distribution, slightly skewed to the right, centered around 63 years. This indicates that the majority of individuals of countries have a moderate to high number of healthy life years, typically between 60 and 65 years, with only a few cases enjoying notably longer healthy lives (up to 75 years). The moderate spread implies limited variation in health outcomes, suggesting a relatively homogeneous and overall favorable health situation.

### 5.5 Table of summary statistics

| | Variables | Mean | SD | Min | Max |
|---|---|---|---|---|---|
| inequality_income_dist | inequality_income_dist | 4.785417 | 1.136924 | 3.03 | 8.32 |
| healthy_life_years | healthy_life_years | 62.288179 | 4.656360 | 51.40 | 73.60 |

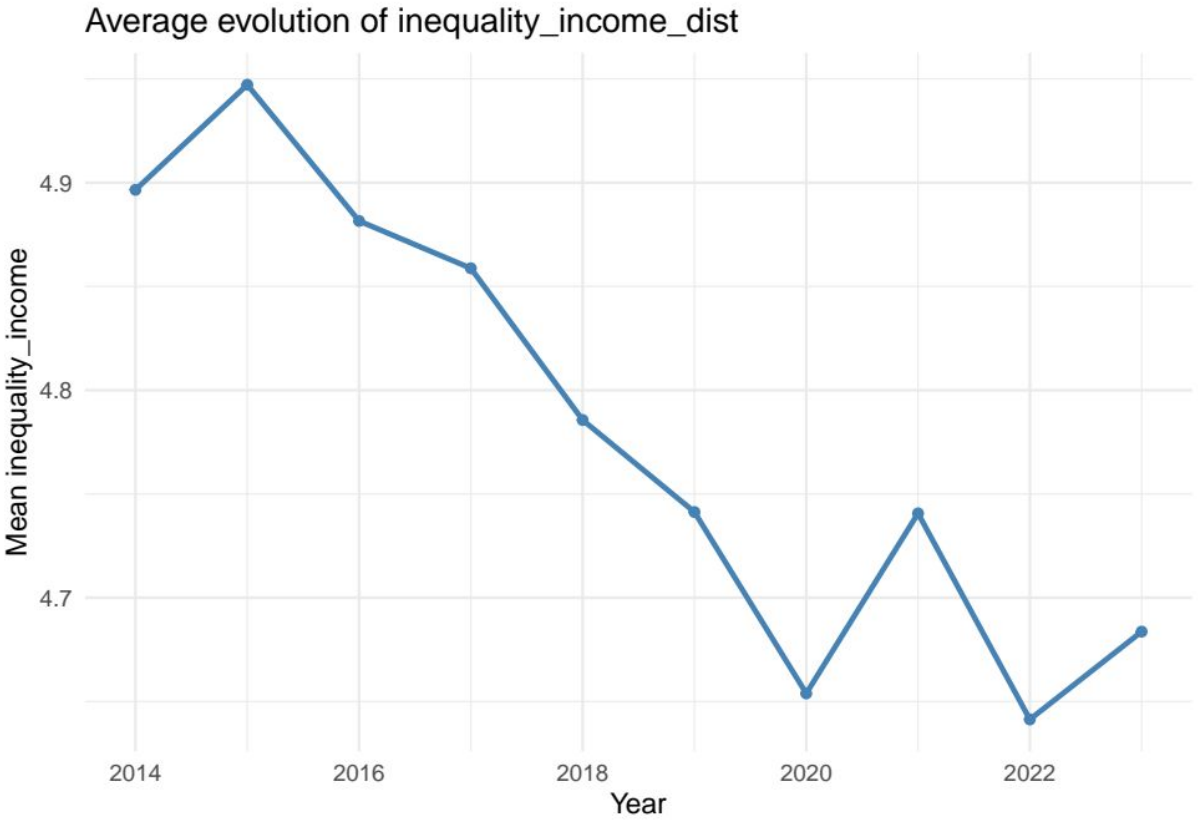
### 5.6 Table of summary statistics per country

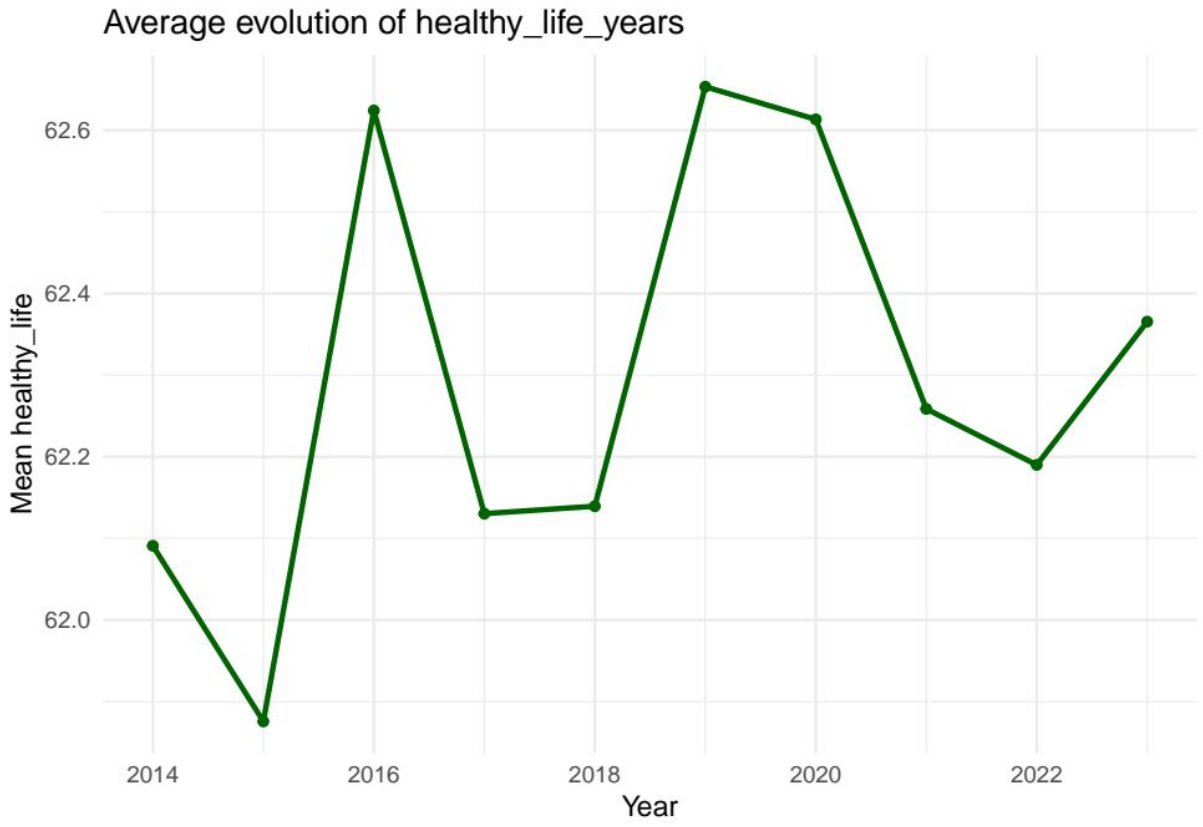
| country | mean_X | sd_X | mean_Y |
|---|---|---|---|
| Bulgaria | 7.497000 | 0.54542644 | 65.37000 |
| Romania | 6.905000 | 0.71694181 | 59.18000 |
| Lithuania | 6.642000 | 0.52253230 | 58.29000 |
| Latvia | 6.424000 | 0.19477622 | 53.09000 |
| Spain | 6.193000 | 0.49766231 | 65.49000 |

```
                                    Italy 5.841000 0.27245591 66.64000
                                   Greece 5.775000 0.58761760 65.41000
                                 Portugal 5.563000 0.41989549 58.50000
                                  Estonia 5.464000 0.50482560 56.42000
                           United Kingdom 5.288000 0.23080295 62.86000
European Union - 27 countries (from 2020) 5.000000 0.17776389 63.39000
                                  Croatia 4.850000 0.18110770 58.46000
                                  Germany 4.766000 0.26742392 64.23000
                               Luxembourg 4.726000 0.34506682 61.57778
                              Switzerland 4.666000 0.20145030 59.87000
                                   Cyprus 4.595000 0.42913997 64.70000
                                    Malta 4.486000 0.42893149 71.81000
                                   Poland 4.384000 0.38123192 62.31000
                                   France 4.375000 0.14001984 64.15000
                                  Hungary 4.273000 0.08380533 61.30000
                                   Sweden 4.242000 0.20170385 71.13000
                                  Ireland 4.207000 0.39564434 67.57000
                                  Austria 4.141000 0.10343543 58.58000
                                  Denmark 4.065000 0.06416126 58.66000
                              Netherlands 3.940000 0.10208929 59.97000
                                   Norway 3.741000 0.20217979 69.27778
                                  Belgium 3.668000 0.18041926 63.76000
                                  Finland 3.647000 0.08446564 57.89000
                                  Czechia 3.424000 0.07275530 62.39000
                                 Slovenia 3.423000 0.14952146 61.03000
                                 Slovakia 3.394000 0.29997037 56.29000
                                  Iceland 3.348571 0.15301727 65.98000
European Union - 28 countries (2013-2020)      NaN          NA 63.10000
     sd_Y obs
1.6680328  10
0.6545567  10
1.4903020  10
0.8047774  10
3.0145757  10
2.3186203  10
0.9993887  10
1.0603982  10
1.6012495  10
1.0163661  10
0.9608445  10
1.5428689  10
3.2639615  10
1.5706510  10
1.2587913  10
1.9223828  10
1.6127616  10
0.5506562  10
0.8058812  10
1.5025904  10
2.8906170  10
1.3589620  10
1.8054239  10
1.8833776  10
0.9730251  10
```

```
1.9816519  10
0.5834762  10
1.4517805  10
0.8582281  10
4.5614447  10
0.8786985  10
3.0865839  10
0.9617692  10
```

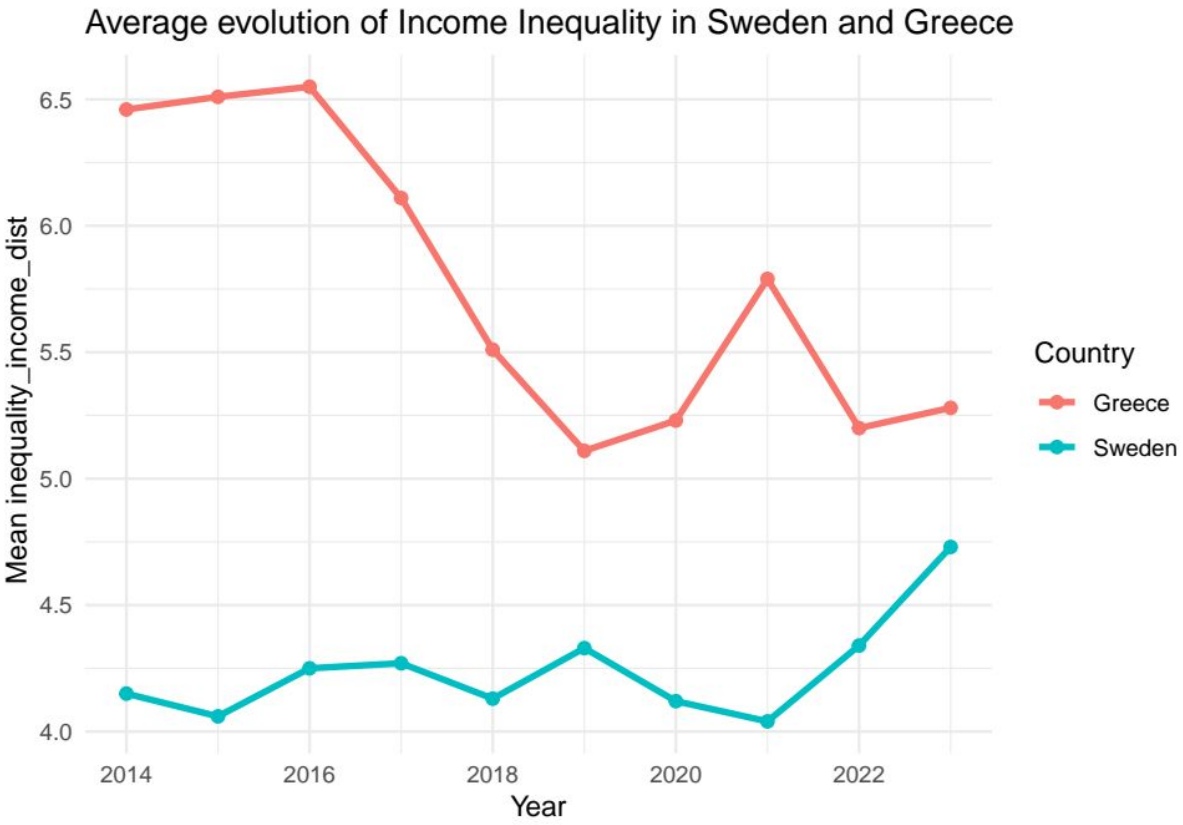
### 5.7 Average evolution of inequality_income_dist and healthy_life_years over time

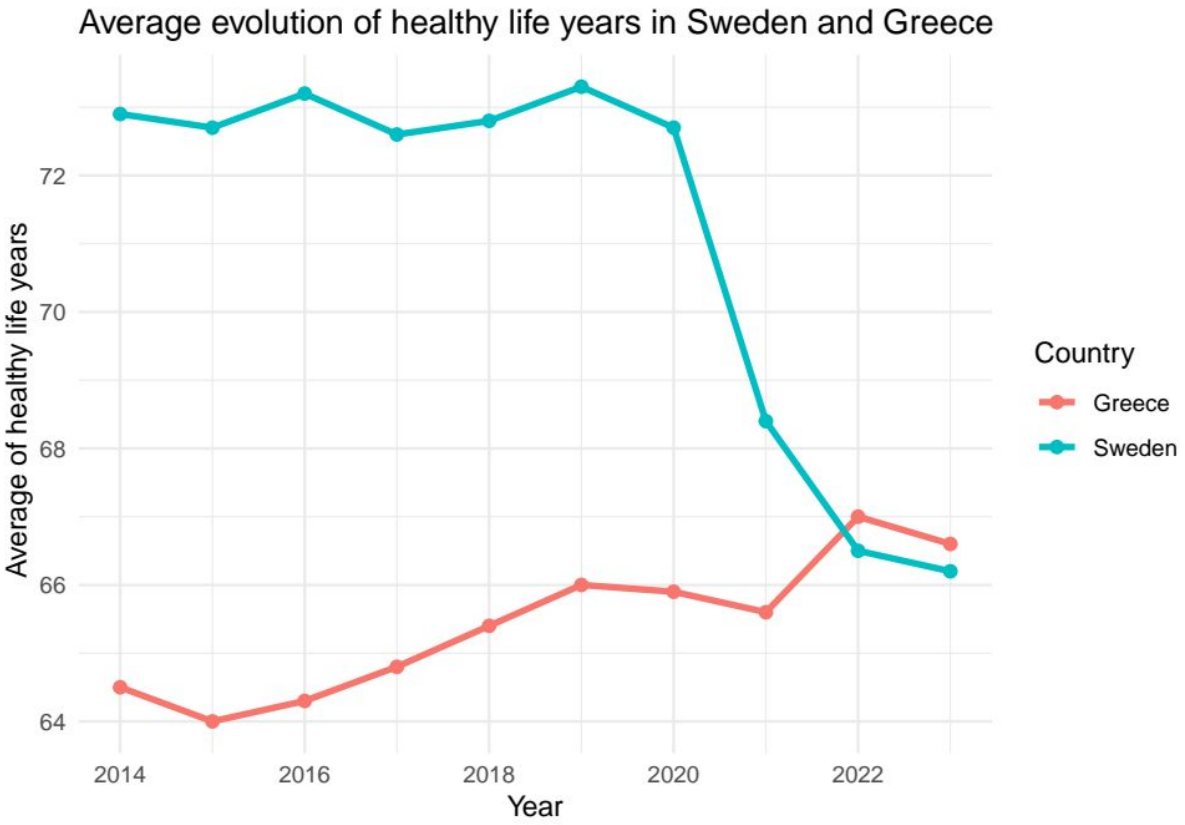





### 5.8 The average evolution of inequality_income and healthy_years in Sweden and Greece.

```
‘summarise()‘ has grouped output by ’country’. You can override using the
‘.groups‘ argument.
```


Average evolution of Income Inequality in Sweden and Greece
Mean inequality_income_dist
6.5
6.0
5.5
5.0
4.5
4.0
2014
2016
2018
2020
2022
Year
Country
Greece
Sweden




Sweden and Greece make an interesting comparison because they represent two contrasting socio-economic models: Sweden is characterized by low inequality income distribution and strong welfare redistribution, while Greece has historically experienced higher inequality, notably intensified after the 2008 financial crisis. In terms of wealth, Sweden generally maintains high and stable accumulation, whereas Greece has faced more volatility and losses during periods of economic instability. Comparing these two countries therefore provides valuable insight into how different economic structures and political strategies; strong welfare protections versus financial vulnerability; shape the evolution of inequality (X) and wealth (Y) over time.

### 5.9 Ranking countries by the change in X and Y between the first and last year

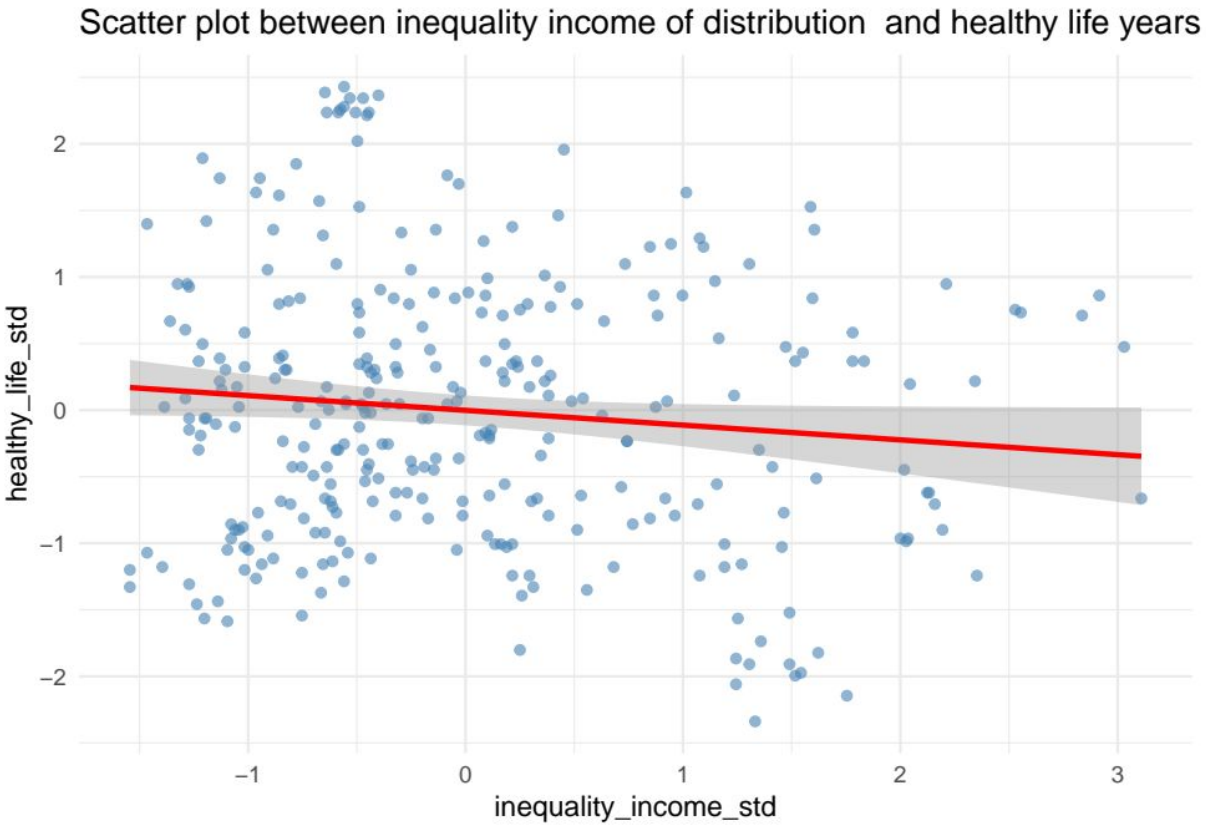



# 6 Relationship between Inequality of income distribution and healthy life years

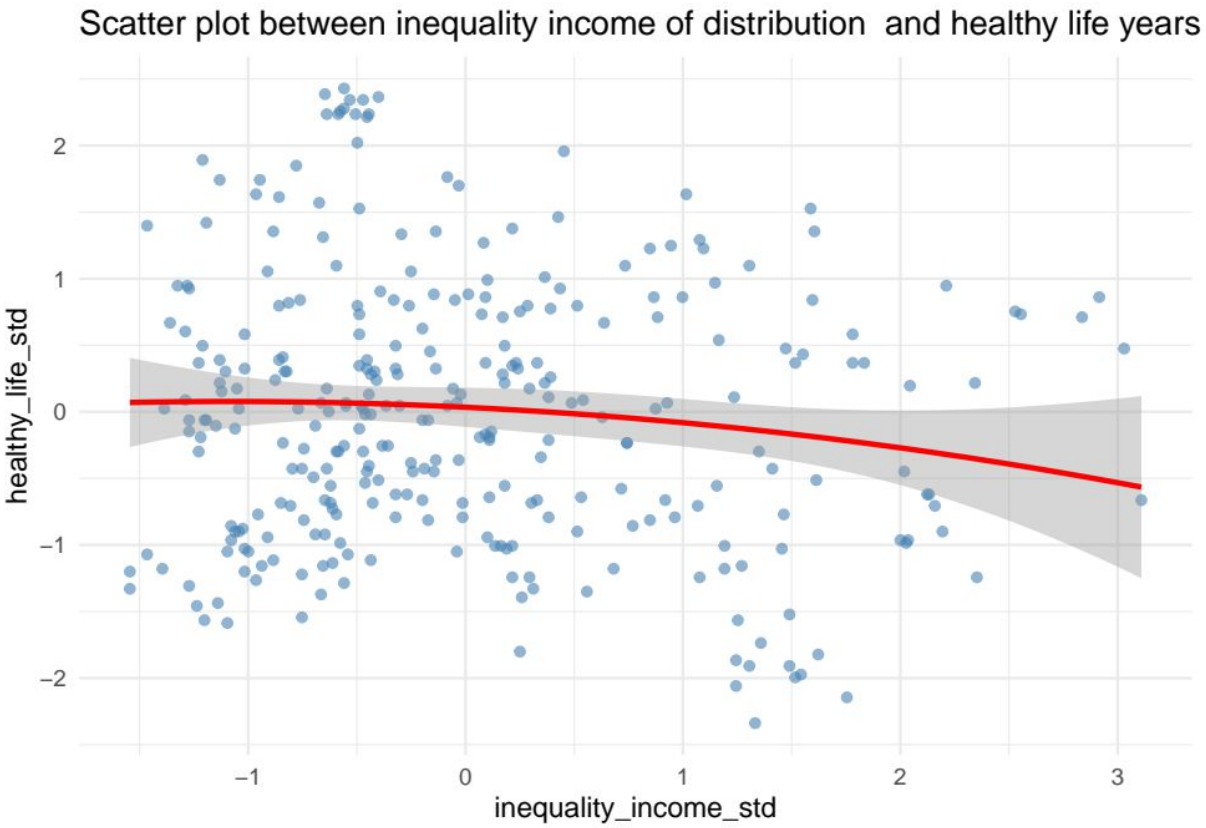
## 6.1 Scatter plot for the relationship between Y and X with linear fit

In order to improve the comparability of the variables, we standardized inequality_income_dist and healthy_life_years. Since these two indicators are not expressed using the same unit of measurement, it was initially difficult to clearly observe the relationship between them. Standardization places both variables on the same scale, with a mean of 0 and a standard deviation of 1, which facilitates graphical visualization and the interpretation of regression coefficients. From this point onward, all analyses in this section are performed using the standardized values of X and Y, allowing for a clearer and more consistent examination of their relationship and the relative magnitude of their variations.

```
Warning: Using one column matrices in ‘filter()‘ was deprecated in dplyr 1.1.0.
i Please use one dimensional logical vectors instead.
This warning is displayed once every 8 hours.
Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
generated.
```


Scatter plot between inequality income of distribution and healthy life years
healthy_life_std
inequality_income_std
2
1
0
−1
−2
−1
0
1
2
3

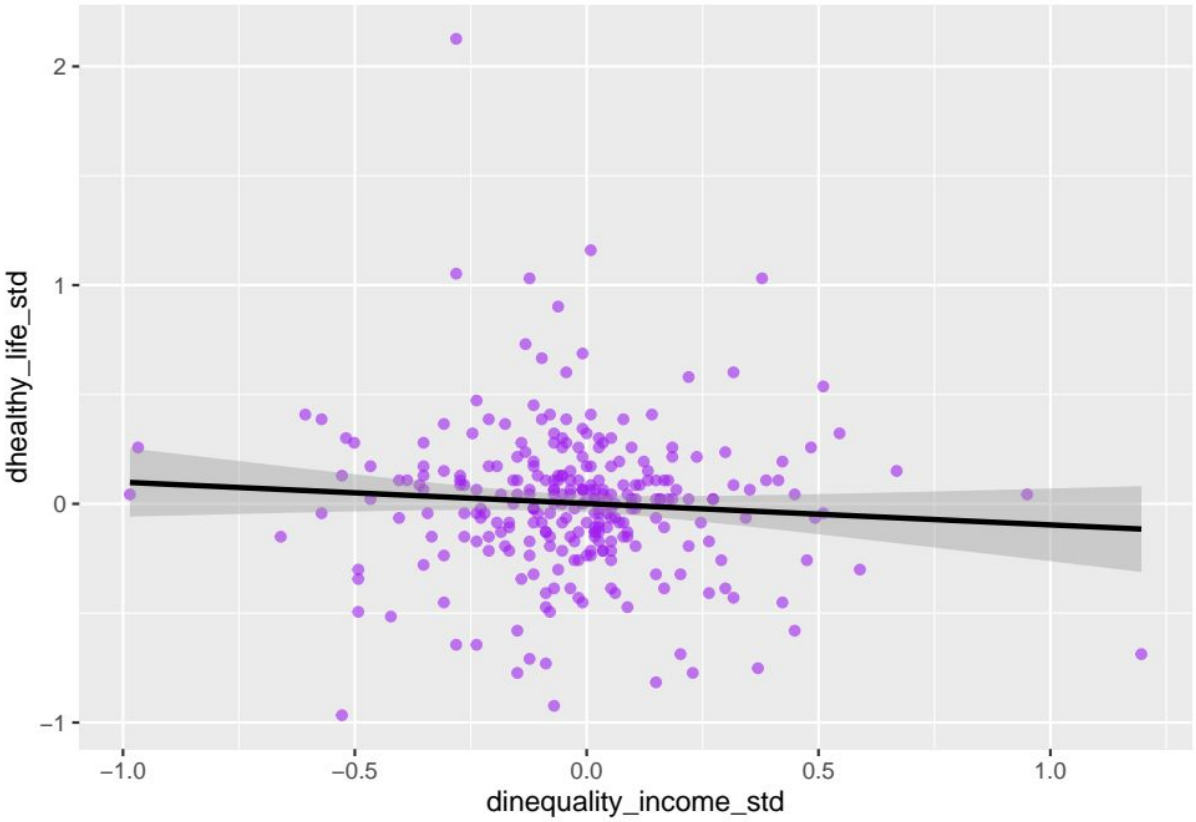
### 6.2 Scatter plot for the relationship between Y and X with a quadratic fit




### 6.3 Compute within-country year-to-year changes

```
# A tibble: 5 x 8
   year country healthy_life_years inequality_income_dist inequality_income_st~1
  <dbl> <fct>                <dbl>                  <dbl>                   <dbl>
1  2015 Austria               58                     4.05                  -0.647
2  2016 Austria               57                     4.09                  -0.612
3  2017 Austria               57.1                   4.29                  -0.436
4  2018 Austria               56.9                   4.04                  -0.656
5  2019 Austria               57.3                   4.17                  -0.541
# i abbreviated name: 1: inequality_income_std[,1]
# i 3 more variables: healthy_life_std <dbl[,1]>,
#   dinequality_income_std <dbl[,1]>, dhealthy_life_std <dbl[,1]>
```

### 6.4 Scatter plot between dinequality_income_std and dhealthy_life_std with linear fit




```
NULL
```

### 6.5 Regress Y on X

#### 6.5.1 Estimation table of model_1

```
Effect of inequalityon healthy
================================================

                       simple Regression results
------------------------------------------------
inequality_income_std           -0.111*
                                (0.057)
Constant                        -0.002
                                (0.057)
------------------------------------------------
Observations                      308
R2                               0.012
================================================
```

## 6.6 Regress Y on X controlling for country fixed effects.

### 6.6.1 Estimation table of model_1 vs model_2

| | Model 1 | Model 2 |
|---|---|---|
| (Intercept) | -0.00 | |
| | (0.06) | |
| inequality_income_std | -0.11 * | -0.19 ** |
| | (0.06) | (0.08) |
| factor(country)Austria | | -0.91 *** |
| | | (0.13) |
| factor(country)Belgium | | 0.13 |
| | | (0.15) |
| factor(country)Bulgaria | | 1.12 *** |
| | | (0.23) |
| factor(country)Croatia | | -0.81 *** |
| | | (0.12) |
| factor(country)Cyprus | | 0.49 *** |
| | | (0.12) |
| factor(country)Czechia | | -0.21 |
| | | (0.16) |
| factor(country)Denmark | | -0.90 *** |
| | | (0.13) |
| factor(country)Estonia | | -1.15 *** |
| | | (0.13) |
| factor(country)European Union - 27 countries (from 2020) | | 0.27 ** |
| | | (0.12) |
| factor(country)Finland | | -1.14 *** |
| | | (0.15) |
| factor(country)France | | 0.33 *** |
| | | (0.13) |
| factor(country)Germany | | 0.41 *** |
| | | (0.12) |
| factor(country)Greece | | 0.84 *** |
| | | (0.14) |
| factor(country)Hungary | | -0.30 ** |
| | | (0.13) |
| factor(country)Iceland | | 0.55 *** |
| | | (0.20) |
| factor(country)Ireland | | 1.04 *** |
| | | (0.13) |
| factor(country)Italy | | 1.11 *** |
| | | (0.14) |
| factor(country)Latvia | | -1.70 *** |
| | | (0.17) |
| factor(country)Lithuania | | -0.54 *** |
| | | (0.18) |
| factor(country)Luxembourg | | -0.16 |
| | | (0.13) |
| factor(country)Malta | | 1.99 *** |
| | | (0.13) |

```
factor(country)Netherlands                                    -0.64 ***
                                                              (0.14)
factor(country)Norway                                          1.32 ***
                                                              (0.15)
factor(country)Poland                                         -0.06
                                                              (0.13)
factor(country)Portugal                                       -0.68 ***
                                                              (0.13)
factor(country)Romania                                        -0.31
                                                              (0.19)
factor(country)Slovakia                                       -1.52 ***
                                                              (0.16)
factor(country)Slovenia                                       -0.50 ***
                                                              (0.16)
factor(country)Spain                                           0.93 ***
                                                              (0.16)
factor(country)Sweden                                          1.81 ***
                                                              (0.13)
factor(country)Switzerland                                    -0.54 ***
                                                              (0.12)
factor(country)United Kingdom                                  0.21
                                                              (0.18)
-------------------------------------------------------------------------------
R^2                                            0.01            0.87
Adj. R^2                                       0.01            0.85
Num. obs.                                    308             308
===============================================================================
*** p < 0.01; ** p < 0.05; * p < 0.1
```

## 6.7 Regress Y on X controlling for year and country fixed effects.

### 6.7.1 Estimation table of model_1 vs model_2 vs model_3

```
=============================================================================================
                                                 Model 1    Model 2      Model 3
---------------------------------------------------------------------------------------------
(Intercept)                                      -0.00
                                                 (0.06)
inequality_income_std                            -0.11 *    -0.19 **     -0.16 *
                                                 (0.06)     (0.08)       (0.08)
factor(country)Austria                                      -0.91 ***
                                                            (0.13)
factor(country)Belgium                                       0.13         1.04 ***
                                                            (0.15)       (0.18)
factor(country)Bulgaria                                      1.12 ***     1.94 ***
                                                            (0.23)       (0.31)
factor(country)Croatia                                      -0.81 ***     0.08
                                                            (0.12)       (0.18)
factor(country)Cyprus                                        0.49 ***     1.38 ***
                                                            (0.12)       (0.18)
factor(country)Czechia                                      -0.21         0.72 ***
                                                            (0.16)       (0.18)
```

| | | |
|---|---|---|
| factor(country)Denmark | -0.90 *** | 0.01 |
| | (0.13) | (0.18) |
| factor(country)Estonia | -1.15 *** | -0.27 |
| | (0.13) | (0.20) |
| factor(country)European Union - 27 countries (from 2020) | 0.27 ** | 1.16 *** |
| | (0.12) | (0.19) |
| factor(country)Finland | -1.14 *** | -0.22 |
| | (0.15) | (0.18) |
| factor(country)France | 0.33 *** | 1.23 *** |
| | (0.13) | (0.18) |
| factor(country)Germany | 0.41 *** | 1.30 *** |
| | (0.12) | (0.18) |
| factor(country)Greece | 0.84 *** | 1.70 *** |
| | (0.14) | (0.21) |
| factor(country)Hungary | -0.30 ** | 0.60 *** |
| | (0.13) | (0.18) |
| factor(country)Iceland | 0.55 *** | 1.51 *** |
| | (0.20) | (0.23) |
| factor(country)Ireland | 1.04 *** | 1.94 *** |
| | (0.13) | (0.18) |
| factor(country)Italy | 1.11 *** | 1.97 *** |
| | (0.14) | (0.22) |
| factor(country)Latvia | -1.70 *** | -0.85 *** |
| | (0.17) | (0.24) |
| factor(country)Lithuania | -0.54 *** | 0.29 |
| | (0.18) | (0.26) |
| factor(country)Luxembourg | -0.16 | 0.73 *** |
| | (0.13) | (0.19) |
| factor(country)Malta | 1.99 *** | 2.89 *** |
| | (0.13) | (0.18) |
| factor(country)Netherlands | -0.64 *** | 0.27 |
| | (0.14) | (0.18) |
| factor(country)Norway | 1.32 *** | 2.24 *** |
| | (0.15) | (0.18) |
| factor(country)Poland | -0.06 | 0.84 *** |
| | (0.13) | (0.18) |
| factor(country)Portugal | -0.68 *** | 0.19 |
| | (0.13) | (0.21) |
| factor(country)Romania | -0.31 | 0.52 * |
| | (0.19) | (0.27) |
| factor(country)Slovakia | -1.52 *** | -0.60 *** |
| | (0.16) | (0.18) |
| factor(country)Slovenia | -0.50 *** | 0.42 ** |
| | (0.16) | (0.18) |
| factor(country)Spain | 0.93 *** | 1.78 *** |
| | (0.16) | (0.23) |
| factor(country)Sweden | 1.81 *** | 2.71 *** |
| | (0.13) | (0.18) |
| factor(country)Switzerland | -0.54 *** | 0.35 * |
| | (0.12) | (0.18) |
| factor(country)United Kingdom | 0.21 | 1.12 *** |
| | (0.18) | (0.23) |
| factor(year)2014 | | -0.93 *** |
| | | (0.15) |

| | | | |
|---|---|---|---|
| factor(year)2015 | | | -0.98 *** |
| | | | (0.14) |
| factor(year)2016 | | | -0.83 *** |
| | | | (0.15) |
| factor(year)2017 | | | -0.94 *** |
| | | | (0.15) |
| factor(year)2018 | | | -0.95 *** |
| | | | (0.15) |
| factor(year)2019 | | | -0.80 *** |
| | | | (0.15) |
| factor(year)2020 | | | -0.82 *** |
| | | | (0.15) |
| factor(year)2021 | | | -0.84 *** |
| | | | (0.15) |
| factor(year)2022 | | | -0.92 *** |
| | | | (0.15) |
| factor(year)2023 | | | -0.88 *** |
| | | | (0.15) |
| R^2 | 0.01 | 0.87 | 0.87 |
| Adj. R^2 | 0.01 | 0.85 | 0.85 |
| Num. obs. | 308 | 308 | 308 |

*** p < 0.01; ** p < 0.05; * p < 0.1

## 6.8 Interpretation

The estimated coefficients show a consistently negative relationship between income inequality and healthy life expectancy. In the simple model, the coefficient is –0.11, indicating that an increase of one standard deviation in inequality is associated with a 0.11 standard-deviation decrease in years of healthy life. When country fixed effects are added, the effect becomes stronger (–0.19) and more statistically significant. This means that when comparing a country to itself over time thereby controlling for structural differences across countries such as development level, healthcare systems, or cultural factors an increase in inequality is associated with a more pronounced loss of healthy life years. In the model that also includes year fixed effects, the coefficient remains negative (–0.16) and significant, confirming the robustness of the relationship even after accounting for time specific shocks, such as the pandemic or economic fluctuations. The coefficients associated with the country and year fixed effects reinforce this interpretation. The country fixed effects show that some countries have, on average, higher healthy life expectancy (positive coefficients, for example, Sweden, Iceland, Malta), while others remain consistently below the baseline (negative coefficients, for example, Bulgaria, Latvia, Slovakia). Finally, the year fixed effects are mostly negative, indicating an overall decline in healthy life years during the study period, regardless of inequality levels. Taken together, these results show not only that income inequality has a direct, statistically significant, and relatively stable effect on health, but also that this effect persists even after controlling for major structural and temporal factors.

## 6.9 Policy implications for economic decision-making

The estimates show that income inequality has a negative, significant, and robust effect on healthy life expectancy. These results suggest that public policies specifically targeting the reduction of income disparities could serve as a key lever for improving population health. First, more progressive redistributive policies (tax reforms, increased progressiveness of income taxes, tax credits for low income households) would help reduce the inequality measured in the models and, according to the estimated coefficients, mitigate part of the decline in healthy life years. Second, strengthening social transfers particularly through housing

benefits, targeted support for vulnerable families, or guaranteed minimum income schemes would directly reduce economic insecurity, a factor often associated with limited access to prevention and healthcare.

At the same time, the country fixed effects show that structural health differences between states are substantial. This supports the need for policies aimed at reducing inequalities in access to health services, notably through greater investment in primary care, prevention, mental health, and hospital infrastructure in regions with weaker health outcomes. Similarly, since the year fixed effects reveal a general downward trend in health over recent years, it is crucial to implement broad strategies that strengthen health resilience: post pandemic public health policies, programs to combat chronic diseases, measures to reduce work related and social stress, and initiatives that improve environmental quality and living conditions. Finally, since an increase of one standard deviation in inequality reduces health by about 0.16 standard deviations in the most comprehensive models, a broad economic policy strategy aimed at reducing inequalities combining taxation, employment, education, and health could generate significant and lasting improvements in healthy life expectancy.